PROGRAMMING DESIGN AND IMPLEMENTATION

Lab 04 Report

File I/O in Java

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Course	Programming Design and Implementation (Java)
Topic	Lab 04: File I/O, 7 exercises + Bonus
Date	November 5, 2026
Repo	https://github.com/starbursttt/PDI-Lab04

Exercise 1: Random numbers to LuckyNumber.csv

Task

> *Generate 500 random numbers and store them in a file named LuckyNumber.csv. Finally, read the data from this file and display it on screen.*

Source code (src/Lab04/Exercise1.java)

HelloWorld.java | Task1.java | Task3.java | Exercise1.java | Exercise2.java

```java
package Lab04;

import java.io.File;


public class Exercise1 {
    public static void main(String[] args) {
        Random r = new Random();

    try {
        FileWriter fw = new FileWriter("LuckyNumber.csv");
        for ( int i =1; i <=500; i++) {
            int num = r.nextInt(10000);
            fw.write(num + "\n");
        }
        fw.close();
        Scanner inp = new Scanner(new File("LuckyNumber.csv"));
        while (inp.hasNextLine()) {
            String line = inp.nextLine();
            System.out.println(line);
        }
        inp.close();
        }catch (IOException e) {
             e.printStackTrace();

        }
    }
}

```

Program output

<terminated> Exercise1 (1)

```
3636
545
3708
3396
1304
6549
7549
6426
1141
6251
6285
8420
3094
1139
7148
9078
5345
4681
4726
5897
4682
6106
2912
7648
6853
6739
558
517
489
1271
9378
293
1327
141
5680
8677
426
4516
```

(.csv files are in github)

Exercise 2: n files of random numbers, max and min

Task

> *Ask the user for a number n. Create n files (file1.csv ... filen.csv) in a folder Ex2LuckyNumber, each storing random numbers between 1 and 100. Read all numbers from the n files into an ArrayList and find the max and min.*

Source code (src/Lab04/Exercise2.java)

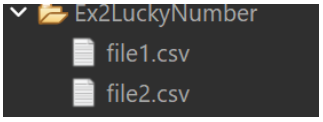
```java
package Lab04;

import java.io.File;
public class Exercise2 {
    public static void main(String[] args) {
        Scanner inp = new Scanner(System.in);
        Random r = new Random();
        System.out.print("Enter n: ");
        int n = Integer.parseInt(inp.nextLine());
        File folder = new File("Ex2LuckyNumber");
        folder.mkdir();
        File[] oldFiles = folder.listFiles();
        for (File f : oldFiles) {
            f.delete();
        }
        try {
            for (int i = 1; i <= n; i++) {
                FileWriter fw = new FileWriter(new File(folder, "file" + i + ".csv"));
                for (int j = 0; j < 10; j++) {
                    fw.write((r.nextInt(100) + 1) + "\n");
                }
                fw.close();
            }
            ArrayList<Integer> numbers = new ArrayList<>();
            for (int i = 1; i <= n; i++) {
                Scanner sc = new Scanner(new File(folder, "file" + i + ".csv"));
                while (sc.hasNextLine()) {
                    numbers.add(Integer.parseInt(sc.nextLine()));
                }
                sc.close();
            }
            int max = numbers.get(0);
            int min = numbers.get(0);
            for (int num : numbers) {
                if (num > max) max = num;
                if (num < min) min = num;
            }
            System.out.println("Total numbers read: " + numbers.size());
            System.out.println("Max: " + max);
            System.out.println("Min: " + min);
        } catch (IOException e) {
            e.printStackTrace();
        }
        inp.close();
    }
}
```

Program output

```
<terminated> Exercise2 (1) [Java Ap
Enter n: 2
Total numbers read: 20
Max: 97
Min: 4
```

Folder contents after running with n = 2



Sample: file1.csv

```
File    Edit    V

96
47
17
70
8
80
79
65
38
97
```

Exercise 3: Write student data to Ex3StudentData.csv

Task

> *Input 3 students (name, ID, gender) of your team members, then store the data in a file named Ex3StudentData.csv.*

Source code (src/Lab04/Exercise3.java)

```java
package Lab04;

import java.io.FileWriter;

public class Exercise3 {

    public static void main(String[] args) {
        try {
            FileWriter fw = new FileWriter("Ex3StudentData.csv");
            Scanner inp = new Scanner(System.in);

            for (int i = 1; i <= 3; i++) {
                System.out.println("Student " + i);

                System.out.print("Input Name: ");
                String name = inp.nextLine();

                System.out.print("Input ID: ");
                String id = inp.nextLine();

                System.out.print("Input Gender: ");
                String gender = inp.nextLine();

                fw.write(name + "," + id + "," + gender + "\n");
            }
            fw.close();
            inp.close();
            System.out.println("Saved to Ex3StudentData.csv!");

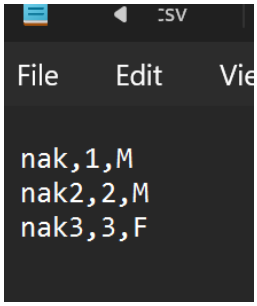
        } catch (IOException e) {

        }
    }
}
```

Program output

```
<terminated> Exercise3 (1) [Java Application
Student 1
Input Name: nak
Input ID: 1
Input Gender: M
Student 2
Input Name: nak2
Input ID: 2
Input Gender: M
Student 3
Input Name: nak3
Input ID: 3
Input Gender: F
Saved to Ex3StudentData.csv!
```

Resulting Ex3StudentData.csv



Exercise 4: Read student data into an ArrayList

Task

> *Read StudentData.csv created in Exercise 3, store the data in an array list, then display the data from the list.*

Source code (src/Lab04/Exercise4.java)

```java
package Lab04;

import java.io.File;

public class Exercise4 {

    public static void main(String[] args) {
        ArrayList<String[]> students = new ArrayList<>();

        try {
            Scanner sc = new Scanner(new File("Ex3StudentData.csv"));
            while (sc.hasNextLine()) {
                String line = sc.nextLine();
                if (line.trim().isEmpty()) {
                    continue;
                }
                students.add(line.split(","));
            }
            sc.close();

            System.out.println("Students loaded: " + students.size());
            for (String[] s : students) {
                System.out.println("Name: " + s[0] + " | ID: " + s[1] + " | Gender: " + s[2]);
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Program output

```
<terminated> Exercise4 [Java Application] C:
Students loaded: 3
Name: nak | ID: 1 | Gender: M
Name: nak2 | ID: 2 | Gender: M
Name: nak3 | ID: 3 | Gender: F
```

Exercise 5: Add, sort (A to Z) and save

Task

> *Load the data in StudentData.csv into an array list. Add 3 more students from your team, sort the data by name (A to Z), and finally write the list back to StudentData.csv.*

Source code (src/Lab04/Exercise5.java)

```java
package Lab04;

import java.io.File;

public class Exercise5 {

    public static void main(String[] args) {
        ArrayList<String[]> students = new ArrayList<>();
        Scanner inp = new Scanner(System.in);

        try {
            Scanner sc = new Scanner(new File("Ex3StudentData.csv"));
            while (sc.hasNextLine()) {
                String line = sc.nextLine();
                if (line.trim().isEmpty()) {
                    continue;
                }
                students.add(line.split(","));
            }
            sc.close();

            for (int i = 1; i <= 3; i++) {
                System.out.println("New Student " + i);
                System.out.print("Input Name: ");
                String name = inp.nextLine();
                System.out.print("Input ID: ");
                String id = inp.nextLine();
                System.out.print("Input Gender: ");
                String gender = inp.nextLine();
                students.add(new String[] { name, id, gender });
            }
            students.sort((a, b) -> a[0].compareToIgnoreCase(b[0]));

            FileWriter fw = new FileWriter("Ex3StudentData.csv");
            for (String[] s : students) {
                fw.write(s[0] + "," + s[1] + "," + s[2] + "\n");
            }
            fw.close();

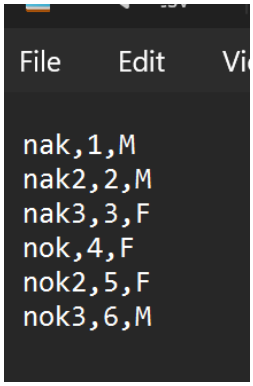
            System.out.println("Saved " + students.size() + " students to Ex3StudentData.csv!");

        } catch (IOException e) {
            e.printStackTrace();
        }
        inp.close();
    }
}
```

Program output

```
<terminated> Exercise5 [Java Application] C:\Users\USER-MSI\
New Student 1
Input Name: nok
Input ID: 4
Input Gender: F
New Student 2
Input Name: nok2
Input ID: 5
Input Gender: F
New Student 3
Input Name: nok3
Input ID: 6
Input Gender: M
Saved 6 students to Ex3StudentData.csv!
```

Ex3StudentData.csv after Exercise 5 (6 students, sorted by name)

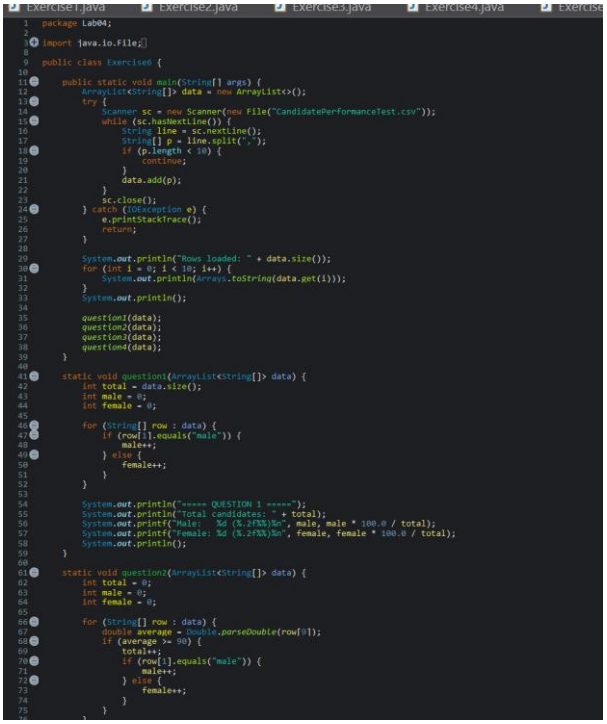


Exercise 6: Analyse the candidate test scores

Task

Read the CSV data and create methods that answer: (1) total candidates and % male/female; (2) candidates with average at least 90 and % male/female; (3) candidates with a score of at least 85 in each subject; (4) students in each group and % male/female.

Source code (src/Lab04/Exercise6.java)



```java
package Lab04;

import java.io.File;

public class Exercise6 {

    public static void main(String[] args) {
        ArrayList<String[]> data = new ArrayList<>();
        try {
            Scanner sc = new Scanner(new File("CandidatePerformanceTest.csv"));
            while (sc.hasNextLine()) {
                String line = sc.nextLine();
                String[] p = line.split(",");
                if (p.length < 10) {
                    continue;
                }
                data.add(p);
            }
            sc.close();
        } catch (IOException e) {
            e.printStackTrace();
            return;
        }

        System.out.println("Rows loaded: " + data.size());
        for (int i = 0; i < 10; i++) {
            System.out.println(Arrays.toString(data.get(i)));
        }
        System.out.println();

        question1(data);
        question2(data);
        question3(data);
        question4(data);
    }

    static void question1(ArrayList<String[]> data) {
        int total = data.size();
        int male = 0;
        int female = 0;

        for (String[] row : data) {
            if (row[1].equals("male")) {
                male++;
            } else {
                female++;
            }
        }

        System.out.println("===== QUESTION 1 =====");
        System.out.println("Total candidates: " + total);
        System.out.printf("Male:   %d (%.2f%%)%n", male, male * 100.0 / total);
        System.out.printf("Female: %d (%.2f%%)%n", female, female * 100.0 / total);
        System.out.println();
    }

    static void question2(ArrayList<String[]> data) {
        int total = 0;
        int male = 0;
        int female = 0;

        for (String[] row : data) {
            double average = Double.parseDouble(row[9]);
            if (average >= 90) {
                total++;
                if (row[1].equals("male")) {
                    male++;
                } else {
                    female++;
                }
            }
        }

        System.out.println("===== QUESTION 2 =====");
        System.out.println("Candidates with average >= 90: " + total);
        System.out.printf("Male:   %d (%.2f%% of them)%n", male, male * 100.0 / total);
        System.out.printf("Female: %d (%.2f%% of them)%n", female, female * 100.0 / total);
        System.out.println();
    }

    static void question3(ArrayList<String[]> data) {
        int math = 0;
        int english = 0;
        int computer = 0;
        int all = 0;

        for (String[] row : data) {
            int m = Integer.parseInt(row[5]);
            int e = Integer.parseInt(row[6]);
            int c = Integer.parseInt(row[7]);

            if (m >= 85) {
                math++;
            }
            if (e >= 85) {
                english++;
            }
            if (c >= 85) {
                computer++;
            }
            if (m >= 85 && e >= 85 && c >= 85) {
                all++;
            }
        }

        System.out.println("===== QUESTION 3 =====");
        System.out.println("Math score >= 85:     " + math);
        System.out.println("English score >= 85:  " + english);
        System.out.println("Computer score >= 85: " + computer);
        System.out.println("85 or more in ALL 3 subjects: " + all);
        System.out.println();
    }

    static void question4(ArrayList<String[]> data) {
        String[] groups = { "group A", "group B", "group C", "group D", "group E" };

        System.out.println("===== QUESTION 4 =====");
        for (String g : groups) {
            int total = 0;
            int male = 0;
            int female = 0;

            for (String[] row : data) {
                if (row[2].equals(g)) {
                    total++;
                    if (row[1].equals("male")) {
                        male++;
                    } else {
                        female++;
                    }
                }
            }

            System.out.println(g + ": " + total + " students");
            System.out.printf("   Male:   %d (%.2f%%)%n", male, male * 100.0 / total);
            System.out.printf("   Female: %d (%.2f%%)%n", female, female * 100.0 / total);
        }
    }
}
```

Program output

```
<terminated> Exercise6 [Java Application] C:\Users\USER-MSI\.p2\pool\plugins\
Rows loaded: 958
[760, male, group A, associate degree, none, 47, 57, 44, 148, 49.33]
[3391, male, group C, some college, none, 76, 78, 75, 229, 76.33]
[3841, male, group B, some college, none, 40, 43, 39, 122, 40.67]
[2632, male, group D, high school, completed, 64, 64, 67, 195, 65.00]
[2779, male, group C, associate degree, none, 58, 54, 52, 164, 54.67]
[1608, male, group D, associate degree, none, 40, 52, 43, 135, 45.00]
[1354, male, group A, some college, completed, 78, 72, 70, 220, 73.33]
[4517, male, group C, high school, none, 88, 89, 86, 263, 87.67]
[9332, male, group C, master degree, completed, 46, 42, 46, 134, 44.67]
[3306, male, group D, high school, none, 66, 69, 63, 198, 66.00]

===== QUESTION 1 =====
Total candidates: 958
Male:   467 (48.75%)
Female: 491 (51.25%)

===== QUESTION 2 =====
Candidates with average >= 90: 48
Male:   14 (29.17% of them)
Female: 34 (70.83% of them)

===== QUESTION 3 =====
Math score >= 85:     112
English score >= 85:  142
Computer score >= 85: 128
85 or more in ALL 3 subjects: 58

===== QUESTION 4 =====
group A: 81 students
   Male:   50 (61.73%)
   Female: 31 (38.27%)
group B: 185 students
   Male:   84 (45.41%)
   Female: 101 (54.59%)
group C: 308 students
   Male:   134 (43.51%)
   Female: 174 (56.49%)
group D: 249 students
   Male:   130 (52.21%)
   Female: 119 (47.79%)
group E: 135 students
   Male:   69 (51.11%)
   Female: 66 (48.89%)
```

Exercise 7: Split candidates into one CSV file per group

Task

Count the students in each group with % male/female, then (a) write the students into separate CSV files by group, and (b) add a summary (total students, % male, % female) at the bottom of each file.

Source code (src/Lab04/Exercise7.java)

Exercise2.java Exercise3.java Exercise4.java Exercise5.java Exercise6.java

```java
package Lab04;

import java.io.File;

public class Exercise7 {
    static String[] groups = { "group A", "group B", "group C", "group D", "group E" };

    public static void main(String[] args) {
        ArrayList<String[]> data = new ArrayList<>();

        try {
            Scanner sc = new Scanner(new File("CandidatePerformanceTest.csv"));
            while (sc.hasNextLine()) {
                String line = sc.nextLine();
                String[] p = line.split(",");
                if (p.length < 10) {
                    continue;
                }
                data.add(p);
            }
            sc.close();
        } catch (IOException e) {
            e.printStackTrace();
            return;
        }

        System.out.println("Rows loaded: " + data.size());
        for (int i = 0; i < 10; i++) {
            System.out.println(Arrays.toString(data.get(i)));
        }
        System.out.println();


        showGroupCounts(data);
        writeGroupFiles(data);
    }

    static void showGroupCounts(ArrayList<String[]> data) {
        System.out.println("===== STUDENTS IN EACH GROUP =====");
        for (String g : groups) {
            int total = 0;
            int male = 0;
            int female = 0;

            for (String[] row : data) {
                if (row[2].equals(g)) {
                    total++;
                    if (row[1].equals("male")) {
                        male++;
                    } else {
                        female++;
                    }
                }
            }

            System.out.println(g + ": " + total + " students");
            System.out.printf("   Male:   %d (%.2f%%)%n", male, male * 100.0 / total);
            System.out.printf("   Female: %d (%.2f%%)%n", female, female * 100.0 / total);
        }
        System.out.println();
    }

    static void writeGroupFiles(ArrayList<String[]> data) {
        try {
            for (String g : groups) {
                String fileName = g.replace(" ", "") + ".csv";
                FileWriter fw = new FileWriter(fileName);

                fw.write("ID,gender,group,Education,test preparation course,Math,English,Computer,Total,Average\n");

                int total = 0;
                int male = 0;
```

```java
                fw.write("ID,gender,group,Education,test preparation course,Math,English,Computer,Total,Average\n");

                int total = 0;
                int male = 0;
                int female = 0;

                for (String[] row : data) {
                    if (row[2].equals(g)) {
                        fw.write(String.join(",", row) + "\n");
                        total++;
                        if (row[1].equals("male")) {
                            male++;
                        } else {
                            female++;
                        }
                    }
                }


                double malePercent = Math.round(male * 10000.0 / total) / 100.0;
                double femalePercent = Math.round(female * 10000.0 / total) / 100.0;

                fw.write("\n");
                fw.write("Total students," + total + "\n");
                fw.write("Male," + male + "," + malePercent + "%\n");
                fw.write("Female," + female + "," + femalePercent + "%\n");

                fw.close();
                System.out.println("Created " + fileName + " (" + total + " students)");
            }
        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

Program output

```
<terminated> Exercise7 [Java Application] C:\Users\USER-MSI\.p2\pool\plugins\c
Rows loaded: 958
[760, male, group A, associate degree, none, 47, 57, 44, 148, 49.33]
[3391, male, group C, some college, none, 76, 78, 75, 229, 76.33]
[3841, male, group B, some college, none, 40, 43, 39, 122, 40.67]
[2632, male, group D, high school, completed, 64, 64, 67, 195, 65.00]
[2779, male, group C, associate degree, none, 58, 54, 52, 164, 54.67]
[1608, male, group D, associate degree, none, 40, 52, 43, 135, 45.00]
[1354, male, group A, some college, completed, 78, 72, 70, 220, 73.33]
[4517, male, group C, high school, none, 88, 89, 86, 263, 87.67]
[9332, male, group C, master degree, completed, 46, 42, 46, 134, 44.67]
[3306, male, group D, high school, none, 66, 69, 63, 198, 66.00]

===== STUDENTS IN EACH GROUP =====
group A: 81 students
   Male:   50 (61.73%)
   Female: 31 (38.27%)
group B: 185 students
   Male:   84 (45.41%)
   Female: 101 (54.59%)
group C: 308 students
   Male:   134 (43.51%)
   Female: 174 (56.49%)
group D: 249 students
   Male:   130 (52.21%)
   Female: 119 (47.79%)
group E: 135 students
   Male:   69 (51.11%)
   Female: 66 (48.89%)

Created groupA.csv (81 students)
Created groupB.csv (185 students)
Created groupC.csv (308 students)
Created groupD.csv (249 students)
Created groupE.csv (135 students)
```

Sample: groupD.csv

```
5688,female,group D,associate degree,completed,74,88,90,252,84.00
1641,female,group D,associate degree,none,59,70,65,194,64.67
8095,female,group D,some college,completed,69,79,81,229,76.33
4077,female,group D,some college,completed,85,86,98,269,89.67
4751,female,group D,associate degree,none,43,60,58,161,53.67
4167,female,group D,some college,none,55,71,69,195,65.00
7293,female,group D,high school,completed,69,77,78,224,74.67
7132,female,group D,high school,none,67,72,74,213,71.00
3432,female,group D,associate degree,none,77,77,73,227,75.67
4968,female,group D,some college,none,79,89,86,254,84.67
3767,female,group D,associate degree,completed,57,78,79,214,71.33
5327,female,group D,some college,none,63,64,67,194,64.67
548,female,group D,associate degree,completed,42,61,58,161,53.67
3918,female,group D,some college,none,98,100,99,297,99.00
7069,female,group D,high school,none,73,92,84,249,83.00
3712,female,group D,some college,completed,52,59,65,176,58.67
3747,female,group D,some high school,none,48,58,54,160,53.33
2118,female,group D,some high school,none,68,71,75,214,71.33
1315,female,group D,associate degree,none,46,56,57,159,53.00
9878,female,group D,some college,none,60,66,70,196,65.33
20,female,group D,some college,none,64,74,75,213,71.00
3237,female,group D,bachelor degree,none,78,90,93,261,87.00
54,female,group D,some high school,completed,69,86,81,236,78.67
8973,female,group D,high school,none,39,52,46,137,45.67
6984,female,group D,some high school,completed,40,65,64,169,56.33
572,female,group D,some high school,none,59,72,80,211,70.33
4783,female,group D,associate degree,none,64,76,74,214,71.33
3483,female,group D,some college,none,69,65,74,208,69.33
568,female,group D,some high school,completed,65,78,82,225,75.00
564,female,group D,bachelor degree,completed,93,100,100,293,97.67
2722,female,group D,some high school,none,67,84,84,235,78.33
5009,female,group D,some college,completed,79,84,91,254,84.67
5782,female,group D,high school,completed,65,61,71,197,65.67
9677,female,group D,master degree,none,78,91,96,265,88.33
2802,female,group D,some college,none,49,58,60,167,55.67
2496,female,group D,some college,completed,75,77,83,235,78.33
5258,female,group D,master degree,none,92,100,100,292,97.33

Total students,249
Male,130,52.21%
Female,119,47.79%
```

Bonus: Serialize and deserialize objects

Task

Write a program that serializes Java objects into CSV format and deserializes them back into objects, handling any necessary data transformations.

Source code (src/Lab04/Bonus.java)

```java
package Lab04;

import java.io.File;

public class Bonus {

    public static void main(String[] args) {
        try {
            ArrayList<Candidate> candidates = readCsv("CandidatePerformanceTest.csv");
            System.out.println("Objects created from the csv: " + candidates.size());

            saveToSer(candidates, "candidates.ser");
            saveToCsv(candidates, "candidates_out.csv");
            System.out.println("Saved candidates.ser and candidates_out.csv");

            ArrayList<Candidate> fromSer = loadFromSer("candidates.ser");
            ArrayList<Candidate> fromCsv = readCsv("candidates_out.csv");
            System.out.println("Objects loaded from candidates.ser:      " + fromSer.size());
            System.out.println("Objects loaded from candidates out.csv:  " + fromCsv.size());


            System.out.println();
            System.out.println("First 3 objects loaded from candidates.ser:");
            for (int i = 0; i < 3; i++) {
                System.out.println(fromSer.get(i));
            }

            int different = 0;
            for (int i = 0; i < candidates.size(); i++) {
                String original = candidates.get(i).toCsv();
                if (!original.equals(fromSer.get(i).toCsv()) || !original.equals(fromCsv.get(i).toCsv())) {
                    different++;
                }
            }
            System.out.println();
            System.out.println("Objects that changed after saving and loading: " + different);

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    static ArrayList<Candidate> readCsv(String fileName) throws IOException {
        ArrayList<Candidate> list = new ArrayList<>();
        Scanner sc = new Scanner(new File(fileName));
        while (sc.hasNextLine()) {
            String line = sc.nextLine();
            if (line.split(",").length < 10) {
                continue;
            }
            list.add(Candidate.fromCsv(line));
        }
        sc.close();
        return list;
    }

    static void saveToCsv(ArrayList<Candidate> list, String fileName) throws IOException {
        FileWriter fw = new FileWriter(fileName);
        for (Candidate c : list) {
            fw.write(c.toCsv() + "\n");
        }
        fw.close();
    }

    static void saveToSer(ArrayList<Candidate> list, String fileName) throws IOException {
        FileOutputStream fileOut = new FileOutputStream(fileName);
        ObjectOutputStream out = new ObjectOutputStream(fileOut);
        out.writeObject(list);
        out.close();
    }


    @SuppressWarnings("unchecked")
    static ArrayList<Candidate> loadFromSer(String fileName) throws IOException, ClassNotFoundException {
        FileInputStream fileIn = new FileInputStream(fileName);
        ObjectInputStream in = new ObjectInputStream(fileIn);
        ArrayList<Candidate> list = (ArrayList<Candidate>) in.readObject();
        in.close();
        return list;
    }
}
```

Supporting class (src/Lab04/Candidate.java)

```java
package Lab04;

import java.io.Serializable;

public class Candidate implements Serializable {
    private static final long serialVersionUID = 1L;

    int id;
    String gender;
    String group;
    String education;
    String testPrep;
    int math;
    int english;
    int computer;
    int total;
    double average;

    public Candidate(int id, String gender, String group, String education, String testPrep,
            int math, int english, int computer, int total, double average) {
        this.id = id;
        this.gender = gender;
        this.group = group;
        this.education = education;
        this.testPrep = testPrep;
        this.math = math;
        this.english = english;
        this.computer = computer;
        this.total = total;
        this.average = average;
    }

    public String toCsv() {
        return id + "," + gender + "," + group + "," + education + "," + testPrep + ","
                + math + "," + english + "," + computer + "," + total + "," + average;
    }

    public static Candidate fromCsv(String line) {
        String[] p = line.split(",");
        return new Candidate(Integer.parseInt(p[0]), p[1], p[2], p[3], p[4],
                Integer.parseInt(p[5]), Integer.parseInt(p[6]), Integer.parseInt(p[7]),
                Integer.parseInt(p[8]), Double.parseDouble(p[9]));
    }

    public String toString() {
        return "Candidate " + id + " | " + gender + " | " + group + " | Total: " + total + " | Average: " + average;
    }
}
```

Program output

